Percentages

The term 'per cent' means 'for every hundred'. Thus, if Rs. 12 is the simple Interest charged for every Rs. 100, then this interest is expressed as '12 per cent' or symbolically 12%. Thus a percentage is merely a ratio that a given number bears to 100.

A fraction can be expressed as a percentage when it is multiplied by 100.

Ex. 1. Express 8/25 as a percentage.

Sol. : $8/25 \times 100 = 32\%$

A given percentage can be expressed as a fraction when it is divided by 100.

Ex. 2. Express $8^1/_3\%$ as a fraction.

Sol. : $8^1/_3\% = 25/3 \times 1/100 = 1/12$

To increase a given quantity by a certain percentage, the given quantity is multiplied by the factor (100 + rate)/100.

Ex. 3. The population of a certain city increased by 6% during a period of 3 years. If at the beginning of the period the population was 8,06,000, what was the population at the end of the period?

Sol. : Population at the end of the period

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$= 8,06,000 \times (100 + 6)/100$
$= 8,06,000 \times 106/100$
$= 8,54,360$

To decrease a given quantity by a certain percentage the given quantity is multiplied by the factor (100-rate)/100.

Ex. 4. The profits of a fruitseller decreased by 8% in 2001 when compared with 2000. If his profits in 2001 were Rs. 36,800, what were his profits in 2000?

Sol. : Let his profits in 2000 be p, then

$p \times (100-8)/100 = 36,800$

$\therefore \quad p = 36,800 \times 100/92$

$= \text{Rs. } 40,000.$

Ex. 5. The annual income of A increased from Rs. 5000 to Rs. 6500 and that of another person B decreased from Rs. 7200 to Rs. 6000, calculate the respective percentages of increase and decrease in the incomes of A and B.

Sol. : A's income increased by Rs. 1500

$\therefore$ Increase in income

$= 1500/5000 \times 100 = 30\%$ increase

B's income decreased by Rs. 1200

$\therefore$ decrease in income

$= 1200/7200 \times 100 = 16^2/_3\%$ decrease.

Ex. 6. At a competitive examination, a candidate 'A' gets 30% of the maximum number of marks and fails by 60 marks while another candidate 'B' by getting 62% of the maximum, gets 36 marks more than are necessary for passing. Find the minimum number of marks necessary for passing.

Sol. : Let the maximum number of marks be 'x' and minimum marks 'y', then

$x \times 30/100 = y - 60$

and $x \times 62/100 = y + 36$

i.e. $3x = 10y - 600$

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$x = (10/3)y - 200$

$\therefore$ $[(10/3)y - 200](62/100) = y + 36$

$(62/30)y - 124 = y + 36$

$(62/30)y - y = 124 + 36$

$(32/30)y = 160$

$32y = 160 \times 30$

$2y = 300$

$y = 150$

Minimum marks required for passing = 150.

Ex. 7. If a man receives on the one-fourth of his capital 3 percent, on two thirds 5 percent and on the remaining capital 11 percent, what percentages does he receive on the whole? [N.D.A.]

Sol. : Let his capital be X, then he receives :

$X/4 \times 3/100 + 2X/3 \times 5/100 +$

$[X - (X/4 + 2X/3)] \times 11/100$

$= 3X/400 + 10X/300 + X/12 \times 11/100$

$= X(3/400 + 10/300 + 11/1200)$

$= X[(9 + 40 + 11)/1200]$

$= X(60/1200)$

$= X \times 5/100$

$\therefore$ He receives 5% on his whole capital.

Ex. 8. A municipality of a certain town earns Rs. 39,900 from property tax charged on all the property situated within its limits. It allows the property owners an allowance at a rate of 5% of the rateable value of the property, before calculating tax. If the rateable value of the property is Rs. 4,20,000 find the rate of property tax.

Sol. : 5% of Rs. 4,20,000 = Rs. 21,000.

This is to be deducted from the value of the property before calculating tax.

$\therefore$ Taxable value of the property

= 4,20,000 – 21,000

= Rs. 3,99,000

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Thus on the rateable value of Rs. 3,99,000, tax of Rs. 39,000 is charged.

$\therefore$ Rate of property tax

$= 39900/399000$

$= 10\%$

Ex. 9. A Zilla Parishad raises Rs. 70,000 per year by way of education tax levied on all the property in its limits. If the rate of taxation is 2 paise in a rupee, find the rateable value of the property.

Sol. : When the tax is Rs. 0.02, value of the property is Re.1

$\therefore$ when the tax collected is Rs. 70,000,

value of the property would be $70000 \times 100/2$

$= 70,000/0.02$

$=$ Rs. 35,00,000

Ex. 10. A Zilla Parishad wishes to construct a new hospital at the cost of Rs. 2,50,000. The amount is raised by taking a loan from a bank at 5% p.a. simple interest. The authorities wish to pay the loan along with the interest after 10 years. This amount is raised by charging the property worth Rs. 9,37,300 in its area, every year. Find the rate of property tax.

Sol. : The total amount to be returned to the bank after 10 years is the capital of Rs. 2,50,000 and interest in years of Rs. 1,25,000. Thus the total amount to be repaid is Rs. 3,75,000.

To repay this after 10 years, the authorities would like to set aside an amount of Rs. 37,500 every year. This is to be collected every year by charging the property worth Rs. 9,37,500.

$\therefore$ Rate of property tax to be levied

$= 37,500/9,37,500$

$= 1/25$

i.e. 4 paise in a rupee or 4%.

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